

PY 765: Homework #4

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Problem 1

Practice with calculations in Fock Space

Assume $a_\sigma^\dagger$ and a_σ are creation and annihilation operators for particles with a CSCO labeled by σ , while $b_\alpha^\dagger$ and b_α are the creation and annihilation operators for the same particles labeled with a CSCO α , such that $\langle 0|b_\alpha a_\sigma^\dagger|0\rangle = U_{\alpha,\sigma}$ is the unitary transformation for the basis change.

(a) Show that

$$\begin{aligned} \langle 0|a_{\sigma_1}a_{\sigma_2}a_{\sigma_3}a_{\sigma'_3}^\dagger a_{\sigma'_2}^\dagger a_{\sigma'_1}^\dagger|0\rangle &= \delta_{\sigma_1,\sigma'_1}\delta_{\sigma_2,\sigma'_2}\delta_{\sigma_3,\sigma'_3} + \delta_{\sigma_1,\sigma'_2}\delta_{\sigma_2,\sigma'_3}\delta_{\sigma_3,\sigma'_1} + \delta_{\sigma_1,\sigma'_3}\delta_{\sigma_2,\sigma'_1}\delta_{\sigma_3,\sigma'_2} \\ &\pm \delta_{\sigma_1,\sigma'_1}\delta_{\sigma_2,\sigma'_3}\delta_{\sigma_3,\sigma'_2} \pm \delta_{\sigma_1,\sigma'_3}\delta_{\sigma_2,\sigma'_2}\delta_{\sigma_3,\sigma'_1} \pm \delta_{\sigma_1,\sigma'_2}\delta_{\sigma_2,\sigma'_1}\delta_{\sigma_3,\sigma'_3} \end{aligned}$$

where the negative signs are for fermions.

(b) For fermions show that

$$\langle 0|b_{\alpha_1}b_{\alpha_2}b_{\alpha_3}a_{\sigma_3}^\dagger a_{\sigma_2}^\dagger a_{\sigma_1}^\dagger|0\rangle = \begin{vmatrix} U_{\alpha_1,\sigma_1} & U_{\alpha_1,\sigma_2} & U_{\alpha_1,\sigma_3} \\ U_{\alpha_2,\sigma_1} & U_{\alpha_2,\sigma_2} & U_{\alpha_2,\sigma_3} \\ U_{\alpha_3,\sigma_1} & U_{\alpha_3,\sigma_2} & U_{\alpha_3,\sigma_3} \end{vmatrix}$$

where the right hand side is the determinant of the 3×3 matrix. This is usually called a Slater determinant in the literature.

(c) Let $A = a_\sigma^\dagger A_{\sigma,\sigma'} a_{\sigma'}$ and $B = a_\sigma^\dagger B_{\sigma,\sigma'} a_{\sigma'}$ where we will assume that the repeated indices σ and σ' are summed up. Note that A and B are matrices made up of complex numbers. Then show that

$$[A, B] = a_\sigma^\dagger [A, B]_{\sigma,\sigma'} a_{\sigma'}$$

for both bosons and fermions.

Problem 2

Consider the Fock space Hamilton operator

$$H_0 = \varepsilon \left(c_1^\dagger c_2 + c_2^\dagger c_1 + c_1^\dagger c_3 + c_3^\dagger c_1 + c_2^\dagger c_3 + c_3^\dagger c_2 \right)$$

where c_i 's and $c_i^\dagger$'s are fermionic annihilation and creation operators. Let $|0\rangle$ be the Fock vacuum. Find all the energy eigenvalues of H_0 . Identify the ground state energy and the corresponding energy eigenstate. Express your answer for the last part in terms of the Fock vacuum and appropriate creation operators.

Solution:

I will select an occupation number basis. I'll label it by annihilation in y and creation in x .

$$H_0 |0\rangle = \epsilon \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Eigenvalues:

$$\begin{aligned} -\lambda(\lambda^2 - 1) - 1(-\lambda - 1) + 1(1 + \lambda) &= 0 \\ \implies \lambda &= \{-1, -1, 2\} \end{aligned}$$

Eigenvectors:

$$\begin{aligned} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} |c_1^\dagger, c_2^\dagger, c_3^\dagger\rangle &= |\vec{0}\rangle \\ \implies z &= x - y \\ \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} |c_1^\dagger, c_2^\dagger, c_3^\dagger\rangle &= |\vec{0}\rangle \\ \implies x &= y = z \end{aligned}$$

then, there's a basis for $\lambda = -1$:

$$b_1^\dagger = A_1(c_1^\dagger - c_3^\dagger), \quad b_2^\dagger = A_2(c_2^\dagger - c_3^\dagger)$$

"Basis" for $\lambda = 2$:

$$b_2^\dagger = A_3(c_1^\dagger + c_2^\dagger + c_3^\dagger)$$

Where I've left these non-normalized with A_i being a normalization factor for each.

Eigenenergies:

$$\{-\epsilon, -\epsilon, 2\epsilon\}$$

Ground state being $-\epsilon$.

Problem 3

* **Hardcore Bosons.** Consider the above problem but now replace c_i 's and $c_i^\dagger$'s with a_i 's and $a_i^\dagger$'s that are bosonic annihilation and creation operators. Also assume the Hamilton operator is modified to $H = H_0 + H_{\text{int}}$ where

$$H_{\text{int}} = U \left(n_1(n_1 - 1) + n_2(n_2 - 1) + n_3(n_3 - 1) \right),$$

and $n_i = a_i^\dagger a_i$ is the bosonic number operator. In the limit $U \rightarrow \infty$ we obtain the hard core boson Hamiltonian. For hardcore bosons, find all finite energy eigenvalues of H . Hint: Once you impose the hardcore boson constraint in the Hilbert space, you can ignore H_{int} and $H = H_0$ like in the fermionic case.

Solution:

Jordan-Wigner mapping:

$$a_i^\dagger = c_i^\dagger \prod_{j < i} (-1)^{n_j}, \quad a_i = \left(\prod_{j < i} (-1)^{n_j} \right) c_i$$

$$H_{\text{eff}} = \varepsilon \sum_{i \neq j} c_i^\dagger c_j = 2\varepsilon c_1^\dagger c_1 - \varepsilon c_2^\dagger c_2 - \varepsilon c_3^\dagger c_3$$

Where these chained $c_i^\dagger c_i = n_i$ (i.e. it's the number operator). The linear combination of all eigenenergies for each occupation number (being 0 or 1) are the new eigenenergies of this system under the JW transformation.

$$E = 2\varepsilon n_1 - \varepsilon n_2 - \varepsilon n_3$$

$$0, 0, 0 \implies E = 0$$

$$1, 0, 0 \implies E = 2\varepsilon$$

$$0, 1, 0 \implies E = -\varepsilon$$

$$0, 0, 1 \implies E = -\varepsilon$$

$$1, 1, 0 \implies E = \varepsilon$$

$$1, 0, 1 \implies E = \varepsilon$$

$$0, 1, 1 \implies E = -2\varepsilon$$

$$1, 1, 1 \implies E = 0$$

Where this new ground state is in the 2-particle basis, -2ε .

Problem 4

Note that the Hilbert space in problems 2 and 3 have the same dimension. Also the Hamiltonians also look the same in the hardcore boson limit. Yet the two problems are different. Explain!